# International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Mid Exam,Autumn-25

**Course Code**: CSE 3631 **Course Title**: Operating System

**Time**: 1 hours 30 minutes **Full Marks**: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

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**[Answer the questions from the followings]**

1. a) What are the main components of a computer system? Briefly describe each with diagram. CO1 3

   b) Define a Real-Time Operating System (RTOS). Briefly differentiate between a Hard Real-Time System and a Soft Real-Time System, providing specific application example for each. CO1 3

   c) Compare and contrast monolithic, layered and microkernel operating system structures in terms of performance, reliability, and design complexity. C02 4

   **or** System calls represent the fundamental interface between a running application (user mode) and the Operating System kernel (kernel mode). Systematically analyze the concept of System Calls, detailing the mechanism of their execution, their necessity,and their classification based on the services they provide.

2. a) The Process Control Block (PCB) serves as the 'identity card' and 'storage container' for every process in an Operating System. Systematically analyze the structure, function, and importance of the PCB in facilitating the core services of the OS, particularly process switching and scheduling. CO2 4

   b) Imagine a busy Football turf with five teams waiting to use the field. C03 6

| **Student:** | **Arrival Time:** | **Burst Time (Initial)** | **Burst Time (Remaining)** |
|---|---|---|---|
| Real Madrid | 12:05PM | 20 minutes | 10 minutes |
| Manchester United | 12:10PM | 10 minutes | 8 minutes |
| Barcelona | 12:15PM | 30 minutes | 9 minutes |
| Bayern Munich | 12:20PM | 5 minutes | 6 minutes |
| Liverpool | 12:25PM | 7minutes | 3 minutes |

i) Schedule the Football turf using Shortest Remaining Time First (SRTF) Scheduling Algorithm. Calculate the average waiting time.
ii) Draw a gantt chart using Round Robin Scheduling Algorithm. Assume that CPU Time Slice: 3 minutes.

3. a) An industrial temperature control system performs these tasks periodically: CO3 6

| Task | Name | Period (ms) | Execution Time (ms) | Deadline (ms) |
|---|---|---|---|---|
| Task A | Temperature Sensing | 30 | 6 | 30 |
| Task B | Pressure Regulation | 60 | 10 | 60 |
| Task C | Display Update | 120 | 20 | 120 |

Illustrate the schedule using Rate Monotonic Scheduling (RMS) and check whether the system meets its deadlines.

b) Semaphores are an essential synchronization tool used by Operating Systems to solve the Critical Section Problem and manage concurrent access to shared resources. Describe the structure and operation of Semaphores and evaluate their strengths and weaknesses as a synchronization mechanism. C01 4

**or** Spooling is a core technique employed by Operating Systems to manage peripheral devices, especially in the context of print services and batch processing. Critically analyze the role, mechanism, and importance of Spooling in modern operating systems.